TOPOSCOUT

GNSS SURVEY PLANNING SYSTEM

SITE INTELLIGENCE REPORT

REPORT SG-V95GVR
Generated July 14, 2026
Data current as of report date

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
38.3459°N 122.3047°W WGS84	**66 ft** NAVD88 · MSL	**CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83)**	**Zone X** 06055C0510FF	**13.0° E** WMM 2026

APN 038050009000 | **ZONING** AP | **ACRES** 0.481 ac

BEST GNSS WINDOW Tue, Jul 14 · 6:00 AM – 10:00 AM | PDOP 0.98 · 33 sv | **NEAREST CORS** P261 · 14.1 mi ✓

REPORT CONTENTS

SCAN FOR KMZ
Site file · Google Earth
↓ Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 14, 2026

ASSESSOR
Napa County Assessor · 1127 First St, Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

SITE & SIGNAL REPORT

COORDINATES	ELEVATION	MAGNETIC DECLINATION	FEMA FLOOD ZONE
38.3459, -122.3047	**66 ft**	**13.0° E**	**X**
WGS84 / NAD83	Above MSL (approx)	Compass reads 13.0° too far W	AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX
4.7

AVG KP
2.4

SOLAR FLUX F10.7
138 sfu

TEC (VTEC) ⚠ DEGRADED
16.9 TECU
-2.9 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Tue, Jul 14	0.98	6:00 AM – 10:00 AM	33	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location — orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

🔥 WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 14, 2026) · Conditions may change — verify before fieldwork

🛰 GPS CONSTELLATION HEALTH

⚠ 1 Satellite Currently Unhealthy — PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

🌿 Land Cover & Multipath Risk

SKY OBSTRUCTION

MULTIPATH RISK

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.9–0.9 across survey window

PDOP remains ≤ 0.9 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

Tuesday, Jul 14

Mainly clear · 💨 12 mph · **GO**

High 99°F · Low 58°F · Wind max 12 mph · Gusts 16 mph · Precip 0.00" · Sunrise 5:57 AM · Sunset 8:32 PM

🌘 Waning Crescent 0%

GNSS: Avg PDOP **0.98** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.93, 33 sats avg)

GNSS / DOP — HOURLY

	PDOP	SATS	RATING
6:00 AM	0.91	33 sv	IDEAL
7:00 AM	1.00	32 sv	GOOD
8:00 AM	0.88	35 sv	IDEAL
9:00 AM	0.93	33 sv	IDEAL
10:00 AM	1.09	30 sv	GOOD
11:00 AM	1.03	29 sv	GOOD
12:00 PM	0.92	31 sv	IDEAL
1:00 PM	1.20	24 sv	GOOD
2:00 PM	0.97	32 sv	IDEAL
3:00 PM	0.93	33 sv	IDEAL
4:00 PM	0.97	33 sv	IDEAL
5:00 PM	0.98	34 sv	IDEAL
6:00 PM	1.03	28 sv	GOOD
7:00 PM	0.92	31 sv	IDEAL

⛅ **GO — GOOD CONDITIONS**

Partly cloudy | Wind: 12.3 mph (gusts 16.1 mph) | Precip: 0" | Temp: 57.8°–98.7°F

⚠️ **FIELD SAFETY WARNINGS**

🌡️ HIGH HEAT 98.7°F — Monitor crew for heat stress

🚁 DRONE WIND ANALYSIS

✅ **GO** Peak: 12.3/16.1 mph S

☀ Sunrise: **5:57 AM** ☾ Sunset: **8:32 PM** fly window clipped to daylight

✅ Best flight window: **6:00 AM – 9:00 PM (15 hrs)**

🔄 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph ⚡=turbulence	DIR
5 AM	5/6 mph	SE
6 AM	5/7 mph	ESE
7 AM	4/5 mph	ESE
8 AM	1/1 mph	S
9 AM	3/3 mph	S
10 AM	4/4 mph	S
11 AM	6/7 mph	S
12 PM	7/7 mph	S
1 PM	8/9 mph	S
2 PM	10/12 mph	SSW
3 PM	12/16 mph	SSW
4 PM	12/16 mph	SSW
5 PM	9/13 mph	WSW
6 PM	6/11 mph	WSW
7 PM	2/13 mph	SW
8 PM	9/12 mph	SE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS A** ► FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
✈ TFR STATUS ✅ **NO TFRs WITHIN 30 MI** Check B4UFLY ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT
GNSS SURVEY OPERATIONS — NAPA COUNTY, CA
Generated: Tuesday, July 14, 2026

==
SITE CONDITIONS SUMMARY
==

Location: 38.3459, -122.3047 (Napa County)
Dominant Land Cover: Dwarf Scrub
Sky Obstruction: Low | Multipath Risk: Low
Recommended Elevation Mask: 13°
Wildfire Status: No active fires detected within reporting range — ALL CLEAR.

Satellite Health Notes: GPS PRN 11 currently unhealthy (receivers will auto-exclude). PRN 3 and PRN 26 have scheduled maintenance windows in subsequent weeks; not applicable to 14 July survey date.

==
SURVEY VERDICT
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GO — Proceed with survey operations.

Primary rationale: Excellent PDOP geometry (avg 0.98, best window 0.93 with 33 satellites), favorable weather (clear skies, low wind, no precipitation), and manageable ionospheric conditions (moderate activity, TEC slightly below normal). Constellation remains robust despite one unhealthy GPS satellite.

==
BEST SURVEY WINDOW
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Tuesday, 14 July 2026: 0600–1000 hrs (local time)

Why: PDOP 0.93 (optimal), 33-satellite constellation (GPS + GLONASS + Galileo + BeiDou), clear skies, minimal wind, and lowest ionospheric delay window. Morning window avoids afternoon convective risk and heat-related field crew fatigue.

==
BEST 4-HOUR STATIC GNSS SESSION
==

0600–1000 hrs, Tuesday, 14 July 2026

Occupation Parameters:
— Session length: 4 hours minimum (recommended: 4–5 hours due to moderate ionospheric activity)
— Elevation mask: 13° (terrain-based)
— Expected PDOP: 0.93 | Expected satellites: 33
— Post-processing baseline: use dual-frequency receivers and apply ionospheric corrections; process with multi-constellation engine (GPS + GLONASS + Galileo + BeiDou) for redundancy.

Ionospheric advisory: TEC 16.9 TECU (slightly below normal). Moderate Kp activity (peak 4.7) warrants standard ionospheric delay mitigation; 4-hour minimum is acceptable; longer occupation provides additional margin.

==
DRONE OPERATIONS ASSESSMENT
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FAVORABLE CONDITIONS — Drone operations approved if required.

— Wind: Max 12 mph (well within limits for most commercial UAS)
— Visibility: Clear skies, excellent
— Interference: Low multipath risk at site; standard RF shielding practices apply
— Airspace: Verify local NOTAM and sectional chart; Napa County airspace generally permissive for Part 107 operations. Coordinate with any nearby airports.

No environmental or weather constraints.

==
FIELD SAFETY WARNINGS
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— Heat: High of 99°F. Ensure adequate crew hydration and shade breaks during morning session.
— Terrain: Dwarf scrub can mask uneven ground; inspect setups for tripod stability before occupation.
— Sun angle: At 0600 hrs, eastern exposure; confirm equipment is not in direct sun during warm-up period.
— Vehicle access: Confirm road conditions and parking in advance; Napa County terrain may have limited access to remote sites.

==
SUMMARY
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Survey conditions on Tuesday, 14 July 2026 are optimal for GNSS static operations. The 0600–1000 hrs window delivers exceptional PDOP geometry (0.93), full multi-constellation availability (33 satellites), clear weather, and manageable ionospheric activity. Site conditions are favorable with low multipath risk and stable terrain. Recommend 4–5 hour static sessions during this window. Field crew should prepare for heat mitigation and verify vehicle access. Proceed with full confidence.

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk
1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor
1127 First St, Napa, CA 94559

CORS STATIONS — WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
NGS ↗	**P261**	HUNTERHILLCN2004	14.1 mi
NGS ↗	**P198**	PETALUMAIRCN2004	17.5 mi
NGS ↗	**OHLN**	OHLONE PARK	23.5 mi
NGS ↗	**P196**	MEACHUMLFLCN2006	24.0 mi
NGS ↗	**CASR**	SANTA ROSA CA	24.9 mi
NGS ↗	**P267**	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS — WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗	**JT9010**	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9018**	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9009**	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9017**	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
Datasheet ↗	**JT9034**	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
Datasheet ↗	**JT9012**	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
Datasheet ↗	**JT9043**	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X — AREA OF MINIMAL FLOOD HAZARD

Panel: **06055C0510FF**
Effective: Sep 29, 2010
View FEMA Map ↗

DATA COLLECTOR IMPORT — NGS BENCHMARKS

↓ Download .txt File

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────────────────────────────────────────────────────────────────────────
NGS BENCHMARK CONTROL POINTS — Copy / Paste to Data Collector
Datum: NAD83  |  Vertical: NAVD88  |  Source: NGS NDE API
────────────────────────────────────────────────────────────────────────
PID      NAME                    LAT           LON           ELEV_FT  ORDER
────────────────────────────────────────────────────────────────────────
JT9010   13                        38.348333    -122.302778       62.7
JT9018   13 E                      38.348333    -122.302778       59.0
JT9009   13                        38.346667    -122.301111       58.7
JT9017   13 D                      38.345000    -122.301111       58.7
JT9034   20 A                      38.345000    -122.309444       68.0
JT9012   13                        38.351667    -122.304444       62.5
JT9043   25 A                      38.340000    -122.302778       61.4
JT9016   13 D                      38.341667    -122.297778       59.7
────────────────────────────────────────────────────────────────────────
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CALIFORNIA RESERVOIRS (DWR/CDEC) — within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.83 ft (-9 ft)	706 ft	36K af	**67%**
Folsom Lake	66.5 mi	447.42 ft (-19 ft)	466 ft	780K af	**80%**
Lake Oroville	93.4 mi	859.6 ft (-40 ft)	900 ft	2,837K af	**80%**
New Melones Lake	100.1 mi	1,019.82 ft (-68 ft)	1,088 ft	1,660K af	**69%**
Black Butte Lake	101.2 mi	—	228 ft	73K af	**54%**

ARMY CORPS RESERVOIRS (USACE) — within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.8 ft	695.01 ft	—	**100%**
Camanche	89 mi	230.2 ft	221.86 ft	—	**104%**
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	**101%**
Farmington Dam-Pool	99 mi	125.3 ft	117.09 ft	—	**107%**
Black Butte-Pool	101 mi	456.8 ft	458.36 ft	—	**100%**
Don Pedro	137 mi	—	N/R	—	**—**

USGS GROUNDWATER WELLS — within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL
004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

⬡ AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa Valley Coastal Range (N100CACSTL) is a coastal alluvial and marine sediment aquifer formed during Quaternary sea-level fluctuations and stream deposition. It consists of unconsolidated to semi-consolidated sand, gravel, silt, and clay that accumulated in valley settings. Water availability at this site is moderate to good, with water tables ranging from shallow (9–15 ft) to moderate depth (55–88 ft), indicating a productive system, though long-term declines suggest increasing stress from extraction.

FORMATION

Name	*Napa Valley Coastal Range Quaternary Alluvial and Marine Deposits*
Age / Rock	*Quaternary (Holocene to Pleistocene); unconsolidated sand, gravel, silt, and clay*
Extent	*600–950 sq mi (est. USGS/DWR)*

DEPTH & RECHARGE

Depth Range	*51–295 ft (5–30 floors)*
Typical Depth	*80–120 ft (8–12 floors)*
Recharge Source	*Precipitation on valley floor + seasonal stream infiltration + lateral underflow from coastal range uplands*
Transit Time	*15–50 years to 100 ft depth*

WATER QUALITY

pH	*est. 6.9–7.6 (est. USGS/DWR)*
Hardness	*est. 180–380 mg/L CaCO3 (est. USGS/DWR)*
TDS	*est. 280–550 mg/L (est. USGS/DWR)*
Notes	*No sample data — typical for formation. Coastal proximity warrants monitoring for saltwater intrusion in lower-elevation wells.*

TREND

Trend	*Decline from 1977–2014 (shallow wells stable; deeper well shows 65.85 ft drawdown over 37 years). Recent 2022 measurement suggests stabilization or partial recovery.*
Source	*USGS NWIS historical data and est. DWR regional trend analysis*

Seasonal Swing	*est. 8–18 ft seasonal swing (est. USGS/DWR)*

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ LOW SUBSIDENCE RISK — No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

⬡ AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU

The North Coast Coastal Terrace aquifer (N100CACSTL) in this region is characterized by well-drained soils and unconsolidated to semi-consolidated alluvial and terrace deposits that have not experienced significant groundwater overdraft. The site's location outside any SGMA-designated critically overdrafted basin, combined with historically adequate groundwater recharge and moderate extraction rates, indicates minimal subsidence risk. Surveyors operating in this area should expect stable vertical datums with no significant elevation changes attributable to aquifer compaction.

SUBSIDENCE ASSESSMENT

Susceptibility	*Low*
Primary Mechanism	*None - consolidated terrain with adequate groundwater balance*
Annual Rate	*<0.5 mm/yr (est.)*
Historic Loss	*None documented*
Aquifer Compaction	*None - aquifer not over-pumped*
SGMA Status	*Not applicable - not in SGMA critically overdrafted basin*

BENCHMARK IMPLICATIONS

Monument Risk	*Low - monuments stable*
Elevation Validity	*Good - no active subsidence*
GNSS Recommendation	*Standard benchmark verification adequate*

SURVEY ACTION

Required Action	*No special action required - proceed with conventional survey methods*
Dataset	*CA DWR SGMA Bulletin 118 2024.1*
Coverage	*California critically overdrafted basins only - full US coverage pending*

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	92 mph
Ground Snow Load	5.8 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	**12 in minimum**

Est. from IECC Zone 3C — AFI data unavailable

Verify with local building official before construction

Source: ASCE 7-22 Hazard Tool · gis.asce.org

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

⬡ AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate stays mild year-round with minimal temperature swings, so your envelope prioritizes air sealing and moderate insulation rather than extreme R-values, but you'll need to design carefully for occasional high winds up to 92 mph—anchor glazing securely and consider strategic window placement on the leeward (east and south) sides while minimizing west-facing glass to reduce afternoon solar gain and wind exposure. The key passive move is deep overhangs and operable windows: use 3–4 foot overhangs on south and west facades to block summer sun while letting winter sun in, and rely on cross-ventilation through operable windows rather than air conditioning, since cool ocean breezes and mild nights let you naturally condition the interior most of the year.

⬡ AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal marine site presents a well-drained gravelly loam foundation with moderate wind exposure and minimal snow load. Seismic parameters are currently unavailable; a geotechnical investigation is recommended to establish site class confirmation and seismic design category before final structural planning.

FOUNDATION

Type	*Conventional spread footing anticipated pending geotechnical confirmation*
Basis	*Well-drained Coombs gravelly loam with 2–5% slopes suggests competent bearing soil; however, seismic site class D and absence of subsidence risk should be confirmed through site investigation to validate allowable bearing pressure and settlement estimates*

STRUCTURAL IMPLICATIONS

Seismic	*Seismic design category cannot be assessed at this planning stage due to null parameters (Ss, S1, SDS, SD1); consult a licensed engineer with final SDC determination to establish if lateral force-resisting system detailing or special inspections may be warranted*
Wind Exposure	*Exposure C (open terrain with some vegetation) anticipated—typical of coastal/agricultural transition zones; verify final exposure category during site reconnaissance*
Wind	*92 mph 3-second gust wind speed under ASCE 7-22 Risk Category II suggested; standard wind load design methodology anticipated—no hurricane or extreme wind zone exposure indicated*
Snow	*5.8 psf ground snow load anticipated for roof design; standard sloped-roof snow shedding and drainage design likely adequate for this IECC Zone 3C marine climate*

INVESTIGATIONS & INSPECTIONS

Geotech Required	*Yes—geotechnical investigation recommended to confirm Site Class D, establish seismic parameters, verify bearing capacity, and assess frost depth (12 in planning estimate for IECC compliance)*

Liquefaction Screen	*Liquefaction screening recommended pending groundwater depth confirmation; well-drained gravelly soils may present low risk, but subsurface exploration advised to rule out relict clay or silt layers*
Special Inspections	*May be required upon finalization of seismic design category; consult engineer once SDC is established*
Code Edition	*ASCE 7-22 / IBC 2021 (planning reference only)*
Site Class Note	*Site Class D assumed for planning purposes based on NEHRP mapping; geotechnical investigation recommended to confirm via boring, SPT, and shear-wave velocity testing*

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes

SERIES: Coombs

DRAINAGE CLASS: Well drained

HYDRO GROUP: C

SOIL ORDER: Alfisols

SUBORDER: Xeralfs

⬡ AI SOIL INTELLIGENCE — CLAUDE HAIKU

The Coombs series consists of well-drained gravelly loam soils formed in alluvial and colluvial materials derived from mixed rock sources on fan terraces and footslopes typical of the inner Coast Ranges. Walking the site, you will encounter moderate gravel content (15–35%) throughout the profile with a clay loam substratum beginning around 20–30 inches depth. Engineers and surveyors should expect good trafficability, straightforward excavation in upper horizons, but should verify in-situ moisture and bearing capacity before foundation design, particularly given the underlying argillic horizon.

Mixed alluvial/colluvial parent material; footslope to fan terrace positions; annual grassland and scattered oak vegetation.

DRAINAGE

Class	*Well drained*
Seasonal Saturation	*None within 60 inches*
Flooding Risk	*Minimal on 2–5% slopes*
Ponding Risk	*None*

ENGINEERING

Bearing Capacity	*2,500–3,500 psf (shallow foundations; subsoil conditions variable)*
Rating	*Fair*
Shrink-Swell	*Moderate — argillic clay loam substratum at 20–30 inches*
Frost Action	*Low*
Cut Suitability	*Good (gravel-rich upper horizons; good drainage)*
Fill Suitability	*Fair (clay loam substratum — moisture control and compaction verification required)*

SEPTIC

Suitability	*Conditional*
Notes	*Percolation test required — clay loam argillic horizon may restrict drainage; site slope and groundwater depth should be verified prior to system design*

HAZARDS

Excavation	*Moderate — 15–35% gravel content; equipment wear expected*
Expansive Clay	*Low to moderate — argillic substratum has moderate shrink-swell potential*
Erosion	*Low on existing gentle slopes; moderate risk if site is graded or unvegetated; gully initiation possible in concentrated drainage corridors*
Other	*None anticipated*

Data sources: California DWR / CDEC · cdec.water.ca.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision
Water levels current as of report run: Jul 14, 2026, 4:57 AM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

🛢 OIL & GAS WELLS
California — within 10mi radius

8	0	7
total	active	plugged

By status:

7 Plugged **1** Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	**Plugged**	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	**Plugged**	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	**Idle**	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	**Plugged**	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation well located 7.2 miles away, which is beyond any actionable threshold for environmental or operational review. You should verify mineral rights status and any historic easements through a title search, but subsurface oil and gas activity does not present a near-term constraint on development.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

⛏ MINES & MINERAL DEPOSITS USGS MRDS — within ~35mi radius

200 total in search area

10 within 5 mi

108 within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	**Occurrence**	Sand/Gravel	1.6 mi
Basalt Rock Co	**Producer**	Clay	2 mi
Cicero Pumice Quarry	**Occurrence**	PUM	2.9 mi
Mount George King Pumice Deposit.	**Occurrence**	PUM	2.9 mi
Basalt Rock Co.	**Occurrence**	DIT	3.1 mi
Mt George Pumice Deposit	**Producer**	PUM	3.1 mi
Wilson Pumice Quarry	**Occurrence**	PUM	3.7 mi
Unknown	**Occurrence**	Mercury	4.4 mi
Silva Pumice Quarry	**Occurrence**	PUM	4.7 mi
Silva Pumice Quarry	**Producer**	PUM	5 mi
Walker Pumice Quarry	**Occurrence**	PUM	5.1 mi
Walker Pumice Quarry	**Producer**	PUM	5.1 mi
Mountain	**Occurrence**	Mercury	5.4 mi
Unknown	**Occurrence**	Mercury	7.6 mi
Unknown	**Occurrence**	Stone/Crushed	7.8 mi
Unnamed Location	**Past Producer**	Sand/Gravel	8.5 mi
Unnamed Location	**Past Producer**	Sand/Gravel	8.5 mi
Unnamed Prospect	**Occurrence**	Chromium	8.8 mi
Unnamed Location	**Occurrence**	Chromium	9 mi
Unknown	**Occurrence**	Chromium	9.1 mi
Bella Oak	**Past Producer**	Mercury	9.4 mi
Unknown	**Occurrence**	MG	9.7 mi
Serres Pit	**Past Producer**	Sand/Gravel	10.4 mi
Unnamed Location	**Unknown**	Sand/Gravel	10.5 mi
Conn Valley	**Occurrence**	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Mineral Resource Site Assessment – 38.3459°N, 122.3047°W Three active mineral producers operate within 5 miles of the project site. Basalt Rock Co (2 mi, clay) and Mt George Pumice Deposit (3.1 mi, pumice) and Silva Pumice Quarry (5 mi, pumice) should be expected to generate heavy truck traffic on access roads during operating hours, potential dust emissions, and noise from crushing or processing operations. No inactive or historical deposits are located within 1 mile, so Phase 1 ESA assessment for legacy mining is not warranted based on proximity alone.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 Endangered

6 Threatened

12 Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	*SCIENTIFIC NAME*	STATUS
California tiger Salamander	*Ambystoma californiense*	**Endangered**
Conservancy fairy shrimp	*Branchinecta conservatio*	**Endangered**
Contra Costa goldfields	*Lasthenia conjugens*	**Endangered**
Soft bird's-beak	*Cordylanthus mollis ssp. mollis*	**Endangered**
Suisun thistle	*Cirsium hydrophilum var. hydrophilum*	**Endangered**
Vernal pool tadpole shrimp	*Lepidurus packardi*	**Endangered**
Alameda whipsnake (=striped racer)	*Masticophis lateralis euryxanthus*	**Threatened**
California red-legged frog	*Rana draytonii*	**Threatened**
Delta smelt	*Hypomesus transpacificus*	**Threatened**
Northern spotted owl	*Strix occidentalis caurina*	**Threatened**
Vernal pool fairy shrimp	*Branchinecta lynchi*	**Threatened**
western snowy plover	*Charadrius nivosus nivosus*	**Threatened**

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Section 7 Implications – Site Review (38.3459, -122.3047) **ESA Section 7 Triggers & Key Species:** Federal nexus (permits, funding, agency action) will trigger mandatory Section 7 consultation given the 12 species with critical habitat within 25 miles. The most planning-relevant species for site-level assessment are California tiger salamander and California red-legged frog (vernal pool/seasonal wetland dependence), vernal pool fairy shrimp and tadpole shrimp (habitat-specific), and Contra Costa goldfields (grassland indicator). Alameda whipsnake presence warrants habitat connectivity review if upland areas are affected. **Recommended Next Step:** Conduct a Phase 1 biological assessment to determine site-specific habitat suitability and occupied/critical habitat overlap, particularly focused on vernal pools, seasonal wetlands, and associated upland refugia. Obtain current USFWS species occurrence data and critical habitat mapping; if proposed activities have federal nexus and overlap critical habitat, prepare for formal Section 7 consultation with USFWS prior to permitting. **For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.**

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

58
Unique Species

5052
Total Observations

2026-07-04
Most Recent

COMMON NAME	*SCIENTIFIC NAME*	LAST OBS
American black nightshade	*Solanum americanum*	2026-05-29
Bearded Clover	*Trifolium barbigerum*	2026-05-15
bird's foot cliffbrake	*Pellaea mucronata*	2026-04-28
Blow Wives	*Achyrachaena mollis*	2026-05-20
blue oak	*Quercus douglasii*	2026-05-09
Bolander's Phacelia	*Phacelia bolanderi*	2026-04-27
box elder	*Acer negundo*	2026-05-18
Braun's Giant Horsetail	*Equisetum telmateia braunii*	2026-06-22
California bay	*Umbellularia californica*	2026-06-24
California beeplant	*Scrophularia californica*	2026-05-20
California black oak	*Quercus kelloggii*	2026-05-15
California buckeye	*Aesculus californica*	2026-06-02
California Dutchman's Pipe	*Aristolochia californica*	2026-05-26
California milkwort	*Rhinotropis californica*	2026-05-15
California mugwort	*Artemisia douglasiana*	2026-06-02
California poppy	*Eschscholzia californica*	2026-06-24
California Wild Rose	*Rosa californica*	2026-05-09
Canyon liveforever	*Dudleya cymosa*	2026-05-30
-	*Castilleja densiflora densiflora*	2026-05-09
clay mariposa lily	*Calochortus argillosus*	2026-05-15
coast live oak	*Quercus agrifolia*	2026-05-13
coastal woodfern	*Dryopteris arguta*	2026-05-20
common selfheal	*Prunella vulgaris*	2026-06-22
common yarrow	*Achillea millefolium*	2026-05-18
Diogenes' lantern	*Calochortus amabilis*	2026-05-15
distant phacelia	*Phacelia distans*	2026-05-15
-	*Downingia concolor concolor*	2026-05-15
Dwarf Brodiaea	*Brodiaea terrestris*	2026-05-15
Elegant Clarkia	*Clarkia unguiculata*	2026-05-16

COMMON NAME	*SCIENTIFIC NAME*	LAST OBS
goldback fern	*Pentagramma triangularis*	2026-05-20
Ithuriel's Spear	*Triteleia laxa*	2026-06-15
maroonspot calicoflower	*Downingia concolor*	2026-05-15
narrowleaf milkweed	*Asclepias fascicularis*	2026-06-25
narrowleaf mule-ears	*Wyethia angustifolia*	2026-05-15
Narrowleaf Willow	*Salix exigua*	2026-06-24
needleleaf navarretia	*Navarretia intertexta*	2026-05-15
northern California black walnut	*Juglans hindsii*	2026-06-24
orange bush monkeyflower	*Diplacus aurantiacus*	2026-05-29
Oregon Ash	*Fraxinus latifolia*	2026-06-02
Pacific Bleeding Heart	*Dicentra formosa*	2026-05-04
Pacific madrone	*Arbutus menziesii*	2026-07-04
Pacific poison oak	*Toxicodendron diversilobum*	2026-06-02
pineapple-weed	*Matricaria discoidea*	2026-05-20
Redwood Sorrel	*Oxalis smalliana*	2026-05-28
small-headed clover	*Trifolium microcephalum*	2026-05-15
Superb Mariposa Lily	*Calochortus superbus*	2026-06-11
thimble clover	*Trifolium microdon*	2026-05-15
tomcat clover	*Trifolium willdenovii*	2026-05-15
Toyon	*Heteromeles arbutifolia*	2026-06-19
trailing blackberry	*Rubus ursinus*	2026-06-24
Tree Clover	*Trifolium ciliolatum*	2026-05-15
valley oak	*Quercus lobata*	2026-06-24
wavy-leafed soap plant	*Chlorogalum pomeridianum*	2026-05-20
western sycamore	*Platanus racemosa*	2026-05-17
white brodiaea	*Triteleia hyacinthina*	2026-05-15
white-tipped clover	*Trifolium variegatum*	2026-05-15
Winecup Clarkia	*Clarkia purpurea*	2026-05-15
Yellow Mariposa Lily	*Calochortus luteus*	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment - Site 38.3459, -122.3047 This site supports a **oak woodland and chaparral mixed community**, characteristic of the inner Coast Ranges of California, with blue oak and California black oak as key canopy species alongside California bay and buckeye. The diverse understory of native forbs, clover species, and wildflowers indicates a Mediterranean climate ecosystem with seasonal water availability. Given the presence of moisture-indicator species like California wild rose, horsetail, and Dudleya (which typically occupy drainage areas and seeps), the site likely experiences seasonal water concentration, suggesting adequate internal drainage but potential for localized wet pockets—a factor critical for grading plans to avoid disrupting natural drainage patterns and seasonal wetland features. Any grading or drainage modifications should carefully preserve existing topography and native vegetation patterns to maintain site hydrology. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

636	155	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants

VENOMOUS SNAKES

SPECIES	OBSERVATIONS
Crotalus oreganus oreganus	636 obs

VENOMOUS SPIDERS

SPECIES	OBSERVATIONS
Latrodectus hesperus	155 obs

STINGING INSECTS

SPECIES	OBSERVATIONS
Dasymutilla aureola	91 obs
Dasymutilla californica	91 obs
Dasymutilla sackenii	91 obs

TOXIC PLANTS

SPECIES	OBSERVATIONS
Toxicodendron diversilobum	1100 obs
Datura wrightii	57 obs
Datura stramonium	57 obs
Datura ferox	57 obs
Conium maculatum	199 obs
Urtica gracilis	83 obs
Urtica gracilis holosericea	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Brief – Napa County Area **Highest priorities:** Northern Pacific rattlesnakes are abundant here (636 observations)—this is your main concern. Poison oak is extremely widespread (1,439 observations) and will contact any exposed skin; black widow spiders and various stinging wasps present secondary hazards. **Practical avoidance:** Wear closed-toe boots, long pants, and long sleeves; watch hand placement when moving rocks or logs where rattlesnakes hide. Stay on cleared paths and avoid brushy areas where poison oak thrives. Learn to identify poison oak's three-leaflet pattern and wash thoroughly after fieldwork. Inspect gear and clothing before leaving site. **Seasonal note:** Rattlesnakes are most active April–October; poison oak causes year-round risk but peaks in spring/summer when oils are potent. Late fall through early spring offers lower snake activity but cold may slow your crew. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3 Beam Paths | **545ft** Closest Beam | **3** Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26 Total Airways | **13** Jet Routes | **2** Victor Airways | **11** Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200	89	64	6	13	6
Total Observations	Total Species	Birds	Mammals	Reptiles	Amphibians

BIRDS (64 species)

COMMON NAME	*Scientific Name*	OBS
California Towhee	*Melozone crissalis*	7
California Quail	*Callipepla californica*	6
Wild Turkey	*Meleagris gallopavo*	6
Snowy Egret	*Egretta thula*	6
Western Bluebird	*Sialia mexicana*	5
Ash-throated Flycatcher	*Myiarchus cinerascens*	4
Great Egret	*Ardea alba*	4
Red-tailed Hawk	*Buteo jamaicensis*	4
Black Phoebe	*Sayornis nigricans*	4
Northern Flicker	*Colaptes auratus*	3
Cooper's Hawk	*Astur cooperii*	3
Western Screech-Owl	*Megascops kennicottii*	3
Turkey Vulture	*Cathartes aura*	3
Mallard	*Anas platyrhynchos*	3
Bushtit	*Psaltriparus minimus*	3
+ 49 additional species		

MAMMALS (6 species)

COMMON NAME	*Scientific Name*	OBS
Coyote	*Canis latrans*	3
Black-tailed Jackrabbit	*Lepus californicus*	2
Mule Deer	*Odocoileus hemionus*	2
Domestic Cat	*Felis catus*	1
Eastern Fox Squirrel	*Sciurus niger*	1
Striped Skunk	*Mephitis mephitis*	1

REPTILES (13 species)

COMMON NAME	*Scientific Name*	OBS
Western Fence Lizard	*Sceloporus occidentalis*	11
Pacific Gopher Snake	*Pituophis catenifer catenifer*	7
Northern Pacific Rattlesnake	*Crotalus oreganus oreganus*	6
California King Snake	*Lampropeltis californiae*	3
Pond Slider	*Trachemys scripta*	2

COMMON NAME	*Scientific Name*	OBS
Western Skink	*Plestiodon skiltonianus*	2
Ring-necked Snake	*Diadophis punctatus*	1
Common Garter Snake	*Thamnophis sirtalis*	1
Western Yellow-bellied Racer	*Coluber constrictor mormon*	1
Pacific Ringneck Snake	*Diadophis punctatus amabilis*	1
Northwestern Pond Turtle	*Actinemys marmorata*	1
Common Sharp-tailed Snake	*Contia tenuis*	1
North American Racer	*Coluber constrictor*	1

AMPHIBIANS (6 species)

COMMON NAME	*Scientific Name*	OBS
Pacific chorus frog	*Pseudacris regilla*	4
Western Toad	*Anaxyrus boreas*	3
American Bullfrog	*Lithobates catesbeianus*	3
Foothill Yellow-legged Frog	*Rana boylii*	3
Rough-skinned Newt	*Taricha granulosa*	1
California Giant Salamander	*Dicamptodon ensatus*	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment **Location: 38.3459, -122.3047** This site supports a diverse wildlife community characteristic of California's oak woodland and grassland transitional habitats with proximate freshwater resources, as evidenced by the presence of both upland species (California Quail, California Towhee, Western Fence Lizard) and obligate aquatic/semi-aquatic species (Snowy Egret, Great Egret, Northwestern Pond Turtle, Pacific Chorus Frog, Foothill Yellow-legged Frog). Several species of regulatory concern are present, including the Foothill Yellow-legged Frog (California Species of Special Concern), Rough-skinned Newt (CESA-listed in some contexts), and multiple native amphibian species indicating relatively intact riparian or wetland habitat within or adjacent to the site. The dominance of native species alongside moderate mammalian predation pressure (coyote, wild turkey predation) and the presence of sensitive aquatic-breeding amphibians suggest the site maintains functional connectivity to natural areas, though the observation of invasive American Bullfrog and domestic cat indicates some anthropogenic disturbance. Overall, this community reflects a partially intact ecosystem with mixed grassland-oak woodland character, seasonal or permanent water features, and moderate habitat sensitivity requiring careful consideration during any site development activities. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

☀ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = **38.3°** (latitude) → **9/12 pitch**

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	← excellent
7/12	30.3°	98.8%	← excellent
8/12	33.7°	99.3%	← excellent
9/12	36.9°	99.8%	← annual optimal
10/12	39.8°	99.8%	← excellent
12/12	45°	99%	← excellent

Going from 5/12 to 6/12 at framing costs ~$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~$108 value/yr). Over 25yr panel life: ~$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	⚠ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	⚠ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr·ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 · ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 · THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 · TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	⚠ verify
50ft	All year	4.8	4.6	7.3	⚠ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
⚠ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†Balanced = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
⚠ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

⬡ AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan You're in a sweet spot for solar—with about 6.6 hours of useful sun each day on average, you'll generate solid power year-round, though summer will be noticeably better than winter (think of it like a savings account that fills faster some months than others). Your roof's angle of roughly 38 degrees is naturally perfect for catching that sun, but here's what most people miss: if your roof is dark, it's absorbing massive amounts of heat on summer afternoons—like parking a black car in the sun—which then radiates into your home and forces your air conditioning to work much harder; a light-colored roof would cut that problem roughly in half. The real sneaky culprit, though, is your west-facing windows: afternoon sun pouring through them in summer creates intense heat gain that's almost double what comes through your south windows, so shading those (with an awning, shade screen, or even a leafy tree) pays immediate dividends in cooling costs. Here's where that 40-foot tree on your south side becomes your friend—it'll shade your home for about 4 hours on summer afternoons when you don't want the heat, but in winter when the sun is lower, it'll let that warmth through for about 5 hours, giving you natural heating for free. Thermal mass is simply the ability of heavy materials (like concrete or adobe) to soak up heat during the day and slowly release it at night, smoothing out temperature swings. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

🧠 SITE INTELLIGENCE SUMMARY AI synthesis — Claude Sonnet

⚙ Technical 👤 Client

Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING **38.34586°N, 122.30466°W | Napa County, California**

Structural Design & Climate Envelope

This Napa County site falls within IECC Zone 3C (Marine) with an Annual Freezing Index of 0°F-days, indicating no freeze-thaw cycling. Wind and snow loads are not reported in the available dataset and should be confirmed through ASCE 7 local design standards and current wind speed mapping; the marine zone classification suggests modest wind exposure relative to inland foothill sites. The absence of documented frost depth reflects the benign freeze environment—footing design will not be constrained by frost penetration. The controlling structural driver cannot be identified from seismic data alone (marked N/A); request site-specific seismic hazard assessment from USGS or California Geological Survey to establish design ground motion parameters, particularly given the Bay Area's active tectonics.

Geotechnical Foundation Profile

The site soils are mapped as Coombs gravelly loam on 2 to 5 percent slopes, classified as Hydrologic Group C with well-drained drainage characteristics. Group C soils have moderately slow infiltration rates and moderate water-holding capacity; well drainage means subsurface water will not pond or create sustained saturation. Gravelly loam composition suggests adequate bearing capacity for conventional shallow foundations, but textural details (fines content, gravel distribution, depth to bedrock) remain unconfirmed and will require site exploration. Recommend a Phase 1 geotechnical investigation including test pits or borings to 6–10 feet to characterize bearing capacity, confirm drainage response during wet winter conditions (standard for Napa), and assess any shallow bedrock or cobble-rich layers that might complicate excavation.

Hazard Assessment

No Quaternary faults are recorded within the search radius, and no subsidence basin data exists for this location. Wind and snow hazard data are not available in the provided dataset. Wildfire risk is not documented but should be independently evaluated given Napa County's fire history; verify defensible space requirements and any local CAL FIRE zone assignments. The absence of seismic fault data does not eliminate seismic risk—obtain a fault map and seismic design category determination from the California Geological Survey or equivalent authority before finalizing structural design parameters.

Mining & Site Context

Three active mineral extraction operations exist within 5 miles: Basalt Rock Co (claystone, 2 miles), Mt George Pumice Deposit (pumice, 3.1 miles), and Silva Pumice Quarry (pumice, 5 miles). Broader MRDS database includes 200 mineral deposits within ~35 miles. Verify that no mining activity or planned extraction occurs on or beneath the subject property and confirm distances from any active pit boundaries. Active mining operations can generate vibration, dust, and traffic; confirm zoning compatibility and obtain any applicable conditional use permits or impact assessments. Eight oil and gas wells exist within 10 miles, all plugged or inactive (nearest is Mobil Oil 7.2 miles, dry hole status). Confirm that no subsurface mineral rights issues or abandoned wellbore proximity affects the development envelope.

Design Priority

Initiate a Phase 1 Geotechnical Investigation paired with a USGS-based seismic hazard and design ground motion assessment to establish both shallow foundation adequacy and lateral load parameters before advancing structural systems or footing design.

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SITE SUMMARY

Coordinates	38.345860°N 122.304660°W
Elevation	66 ft NAVD88 (approx)
Township/Range	T6N, R4W, Mount Diablo Meridian
FEMA Zone	Zone X Panel 06055C0510FF
Nearest BM	JT9010 0.2mi 62.7ftft
Nearest CORS	P261 14.1mi
State Plane	CA 2 (0402)

0 —— 10 —— 20 mi

CORS BM SITE FEMA: Zone X Panel 06055C0510FF

TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

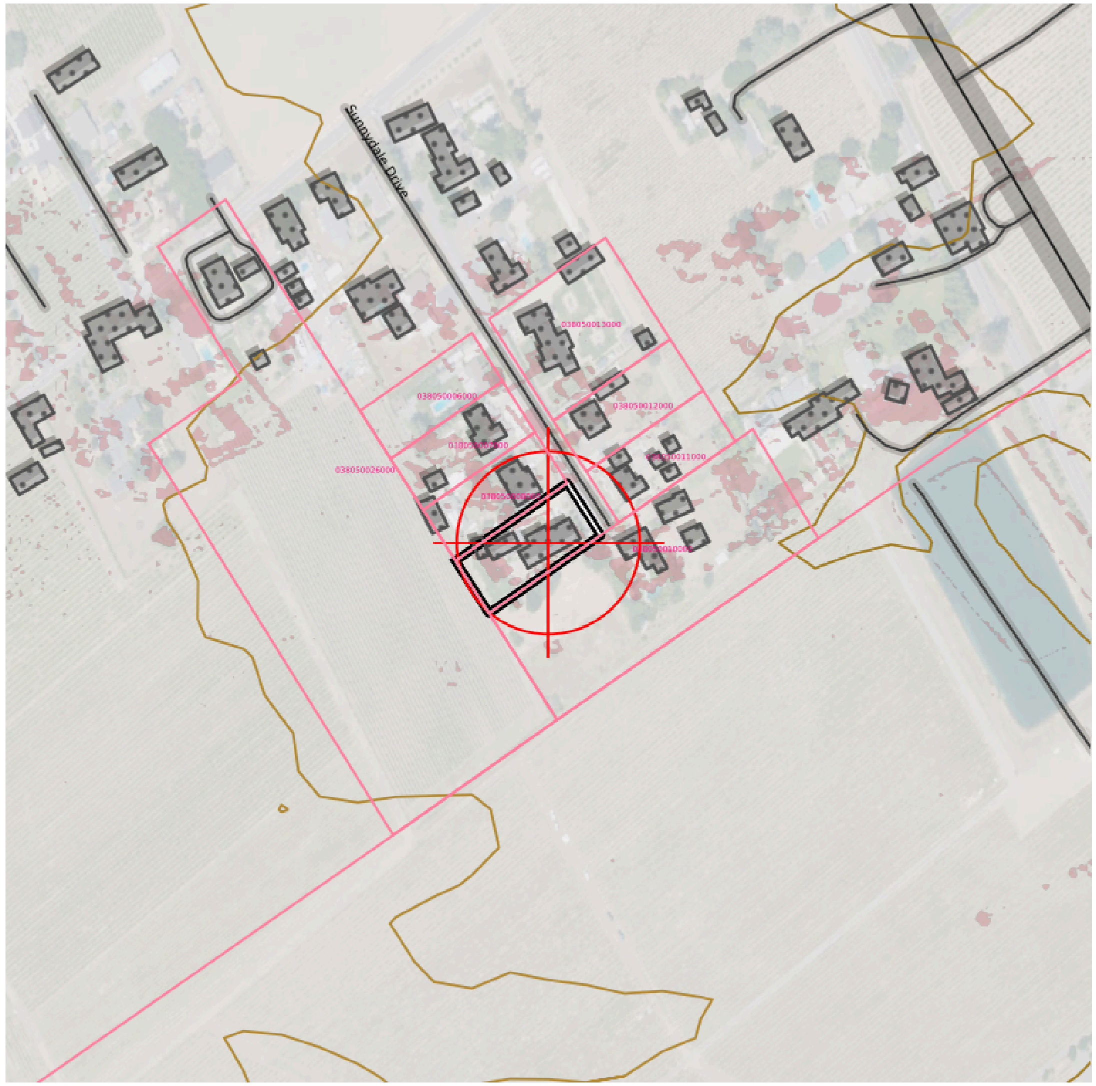
Generated: July 14, 2026
Base: USGS National Map
Generated by TopoScout · outKastdesigns.com

TOPOSCOUT SITE DXF PREVIEW

45 CAD Layers · AutoCAD Compatible · State Plane Coordinates

Need the editable CAD file? Add the DXF at toposcout.com

TOPOSCOUT SITE PLAN PREVIEW



INCLUDED LAYERS

N

0 —— 400 ft State Plane · NAD83 · US Feet

■ Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels ■ County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polylines (all geometry)

SITE AERIAL — 600 ft × 600 ft

38.34586°N, 122.30466°W · Google Aerial

TOPOSCOUT SITE AERIAL IMAGERY

Source: Google Aerial

AREA CONTEXT — 3000 ft × 3000 ft

38.34586°N, 122.30466°W · Google Aerial

⚡ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online – https://trimblegnssplanningonline.com/
Satellite visibility, DOP charts, sky plots

FAA B4UFLY – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – https://www.aloft.ai/
Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp – https://www.swpc.noaa.gov/products/planetary-k-index
Real-time Kp index, geomagnetic storm alerts

BASE STATION & NETWORK

NOAA CORS Network – https://geodesy.noaa.gov/CORS/cors-data.shtml
Continuously operating reference stations, data download

NGS CORS Station Map – https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml
Interactive map of all CORS stations with status

IGS Station Monitor – https://www.igs.org/network
International GNSS Service — global base station health

POST-PROCESSING

NGS OPUS – https://geodesy.noaa.gov/OPUS/index.jsp
Online Positioning User Service — static GNSS processing

NGS OPUS-RS – https://geodesy.noaa.gov/OPUSI/index.jsp
Rapid static processing — sessions as short as 15 minutes

CORS Data Download – https://geodesy.noaa.gov/CORS/cors-data.shtml
Raw RINEX observation files from CORS stations

REFERENCE & CALCULATIONS

NGS Datasheets – https://www.ngs.noaa.gov/cgi-bin/datasheet.prl
Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) – https://geodesy.noaa.gov/NCAT/
NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination – https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml
Declination calculator for any location and date

NGS Geoid Models – https://geodesy.noaa.gov/GEOID/
GEOID18 and current model downloads, GEOID calculator

FEMA & FLOOD

FEMA Flood Map Service Center – https://msc.fema.gov/portal/home
FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate
Current EC form, instructions, LOMA/LOMR submission guidance

DRONE OPERATIONS

FAA Part 107 — Commercial Rules – https://www.faa.gov/uas/commercial_operators
Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – https://faadronezone.faa.gov/
Drone registration, operator certificates, waiver management

FAA B4UFLY – https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly
Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers – https://www.faa.gov/uas/commercial_operators/part_107_waivers
Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – https://www.aloft.ai/
Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

AirNow Fire & Smoke Map
https://www.fire.airnow.gov
Current fire locations, smoke plumes, PM2.5 readings by location

AirNow Interactive Map
https://gispub.epa.gov/airnow
Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

NOAA Smoke Forecast
https://airquality.weather.gov
48-hr smoke concentration forecast, HRRR-Smoke model

Links verified: February 2026 · Check quarterly · TopoScout GNSS Planning System